AXIS PIPE AND TUBE

AXIS PIPE AND TUBE, LLC.
1451 Louis E. Mikulin Rd.
Bryan TX 77807
TEL. (979) 703-6847 FAX(979) 703-6847

MILL TEST CERTIFICATE

Customer:	Sales Order:	Invoice No.:
Industrial Piping Specialists PO Box 681270 Tulsa,OK,74158 918-437-9100	0630005397 **PO:** WP012766	1600051929 **Date of Issue:** 02/14/2025

Certificate No.:	1600051929-2
Product Description:	5L 8.625 0.322 28.58# X52BDRH5BW-20P2
Specification:	5L 46th Edition May 2019

Heat Number	Lot Number	Mat Number	Steel Making / Coil Rolling
B416740	LT25028014	333651	Steel Dynamics Columbus, LLC , Columbus - MS - US / Steel Dynamics Columbus, LLC , Columbus - MS - US

Mechanical Properties

Tensile Test - Strip Specimen Gauge Length 2"								CVN Temp (F) -20								Temp (F) -20										
Loc. / Orien	Size	0.5%EUL Y.S(psi)	U.T.S. Body (psi)	U.T.S. Weld (psi)	Y/T Ratio	E.L. (%)	Hardness (HRB)	Body (ft-lbf) 1	2	3	Avg	Shear Area (%) 1	2	3	Avg	Weld (ft-lbf) 1	2	3	Avg	Shear Area (%) 1	2	3	Avg	Size	Loc. / Orien	Acceptance Criteria (ft-lbf)(AVG)
T180°	1 1/2"	66500	73200	75100	0.91	38.2	82	129	169	155	151	90	100	90	94	129	133	150	138	90	90	90	90	10x6.7 mm	3T90	13.3
								Lateral Expasion (in)				0.074	0.068	UB	0.071	Lateral Expasion (in)				0.085	0.077	0.075	0.079			

Chemical Composition (Max %)

Analysis (H/P)	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Al	Nb	Ti	B	Ca	N	CEpcm (API Max. A.C.)	CE Results
H	0.06	0.210	0.80	0.013	0.002	0.05	0.06	0.020	0.13	0.002	0.024	0.028	0.015	0.0003	0.0020	0.0062	0.25	0.12
P	0.06	0.222	0.80	0.013	0.001	0.05	0.06	0.010	0.13	0.004	0.020	0.025	0.015	0.0003	0.0022	0.0075	0.25	0.12
P	0.06	0.220	0.80	0.013	0.001	0.05	0.06	0.010	0.13	0.004	0.020	0.025	0.015	0.0004	0.0020	0.0076	0.25	0.12

Following Controls / Tests have been satisfactorily performed:
1.- Hydrostatic Testing: 3,900.00 psi Pressure for 5.00 seconds
2: Ultrasonic Tested to N10 - passed
3: Seam Annealed min 1,600 F
4: Flattening Test Passed

Remarks:
1: This material meets:Grades ASTM A53/A53M-22 Grade B
API 5L HFW PSL2 X52M,
2: This product meets the requirements of ASME BPVC.II.A Specification SA-53B-2023
3: Made and Melted in USA
4: Date of Manufacture 02/2025
5: This product is silicon and/or aluminum fully killed.
6: Certificate meets the requirements of ISO 10474:2013 (Inspection Certificate 3.1.B)
7: This material meets the requirements of NACE MR0175-2021 / ISO 15156:2020

Acronyms:

L = Longitudinal	T = Transverse
Y.S. = Yield Strength	U.T.S. = Ultimate Tensile Strength
E.L. = Elongation	H = Heat Analysis
P = Product Analysis	A.C. = Acceptance Criteria
UB = Un-Broken	N/A = Not Applicable

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

Mahir Arora

Quality Manager

Tulsa Tube Bending

Dimensional-Visual Inspection Report

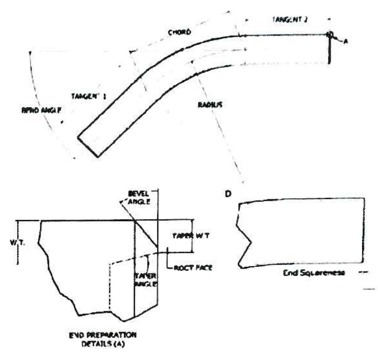
Customer:	DNOW	**Pipe Size (in):**	8.625 x 0.322	**Report No.:**	109149-DV-2-1
Customer PO No.:	662974020	**Grade:**	API 5L X52 PSL 2	**Bend ID No.:**	109149-2-1
TTB Job No.:	26H109149	**Heat No.:**	B416740	**Code of Fabrication:**	ASME B16.49-2023

Physical Dimensions

Dimension	Target	Tolerance	Actual
Tangent 1 (in)	24 MIN	MIN	27.56
Tangent 2 (in)	24 MIN	MIN	30.81
Chord (in)	30.6	n/a	30.8
Angle (deg)	45.0	+/- 0.5 °	45.0
Radius, R (in)	40	+/- 1%	40.2
Performed By:	EB	**Date:**	6/12/2026

End Prep Dimensions

Dimension	Target	Tolerance	Actual	
Bevel Angle (deg)	37.5	+/- 2.5°	38.00	END 1
Root Face (in)	0.06	+/- 0.03 in	0.060	END 1
Min ID	7.981	+/- 0.1 in	7.940	END 1
Max ID	7.981	+/- 0.1 in	7.940	END 1
Max End Sq. (in)	0	+/- 0.09 in	0.060	END 1
Bevel Angle (deg)	37.5	+/- 2.5°	36.00	END 2
Root Face (in)	0.06	+/- 0.03 in	0.060	END 2
Min ID	7.981	+/- 0.1 in	7.940	END 2
Max ID	7.981	+/- 0.1 in	7.940	END 2
Max End Sq. (in)	0	+/- 0.09 in	0.060	END 2
Performed By:	EB	**Date:**	6/12/2026	



Ovality [(O.D. max – O.D. min)/O.D. nominal] x 100

Quadrant	Limit	Min OD	Max OD	% Ovality
End 1	1.0%	8.597	8.640	0.5%
Start of Bend	1.0%	8.502	8.550	0.6%
1/4 of Bend Arc	1.0%	8.536	8.589	0.6%
1/2 of Bend Arc	1.0%	8.529	8.610	0.9%
3/4 of Bend Arc	1.0%	8.517	8.602	1.0%
Stop of Bend	1.0%	8.550	8.594	0.5%
End 2	1.0%	8.605	8.650	0.5%
Performed By:	EB		**Date:**	6/12/2026

Wall Thinning of Extrados

Quadrant	Limit	Nominal Wall Thickness	Extrados Wall Thickness	% Wall Thinning
Start of Bend	10.0%	0.322	0.309	4.0%
1/4 of Bend Arc	10.0%	0.322	0.304	5.6%
1/2 of Bend Arc	10.0%	0.322	0.298	7.5%
3/4 of Bend Arc	10.0%	0.322	0.299	7.1%
Stop of Bend	10.0%	0.322	0.305	5.3%
Performed By:	EB		**Date:**	6/12/2026

ID Gauge Drift/Pig Inspection

Minimum Gauge Diameter =	97.0%	of Nominal ID, or	7.742 in
Gauge Length Required =	7.981 in		

Gauge passes through the entire bend without power equipment assistance

Performed By:	MF	**Date:**	6/15/2026

Visual Inspection

All accessible surfaces found to be free of visible injurious defects.
Bend is stenciled with correct information.

Performed By:	MF	**Date:**	6/15/2026

READINGS:

- [x] MEET ACCEPTANCE CRITERIA
- [ ] DO NOT MEET ACCEPTANCE CRITERIA

Globe X-Ray Services, Inc.
8441 S. Union Ave.
Tulsa, OK 74132
918-446-1696

Certification # 194846

Magnetic Particle Examination Record

Customer: Tulsa Tube Bending Date: 6/12/2026

Job/P.O. No.: 109149 Reference Code/Specification: ASME SEC V

MT Procedure No. GXS IV - C Rev. No. 1/26 Acceptances: ASME SEC VIII Div. I

Magnetizing Equipment: Parker Contour Probe B300 Serial No. 1861 Technique: X Yoke ___ Prods

Current Type: X A/C ___ D/C Magnetic Particle Type: ___ Wet X Dry ___ Fluorescent X Visible

Amperage (Prod Technique Only) ___ Amps/inch of Prod Spacing.

Part/Item Material Type: Carbon Material Thickness: Varies

Lighting Equipment: ___ Fluorescent Light Model # ___ Serial # ___
X White Light using Parker Research Y300 or Y400 Probe light
___ Natural White Light
___ Other: ___

Light Intensity on surface to be examined: 100 Min foot candles N/A μW/cm²

Comments: Test The Entire Extrados (Back Half) of the Bend. From N.A to N,A, Including Weld Seam
Verify Residual Magnetism at Each End is <25 Gauss

Examination Results				
Qty:	Part Identification	Accept	Reject	Indication Description
4	8.625" O.D. X 0.322" Wall, 40" CLR, API5L X52 ERW Pipe HT# B416745 Bend ID # 109149-1-1, 109149-1-2, 109149-1-3, 109149-1-4,	X		None
2	8.625" O.D. X 0.322" Wall, 40" CLR, API5L X52 ERW Pipe HT# B416740 Bend ID # 109149-2-1, 109149-2-2	X		None

Examine: Jaime Vera Qualification Level: II

Tulsa Tube Bending
4192 S. Galveston Ave.
Tulsa, OK 74107
Tel: 918-446-4461
Fax: 918-446-1595
Serving others. Building people. Pursuing excellence.

Certificate of Conformance

Customer: DNOW

Certificate No.: 20260615

Customer PO No.: 662974020

Issue Date: 6/15/2026

TTB Job No.: 26H109149

Material Size, Grade: 8.63 inch OD x 0.322 inch w.t., API-5L-X52 PSL 2 ERW

Quantity of Bends: 6

Material Heat No(s): B416745

Bend ID Nos.: 109149-1-1 109149-1-2 109149-1-3 109149-1-4

Material Heat No(s): B416740

Bend ID Nos.: 109149-2-2 109149-2-1

We hereby certify that the aforementioned induction bends have been inspected and conform to all of the requirements of the Purchase Order. The induction bends have been manufactured in conformance with:

ASME B16.49-2023

*With the following notes and exceptions:

* B16.49 SR15.1 - Bends will be in the as bent condition in accordance with ASME B16.49 SR15.1.
* SR15.8.2 Multiple heats of material used for production bends based on similar chemistries.
* Segmentable per SR15.3, 1% max ovality in the bend area.

Approved By: Nate Wallace

Date: 6/15/2026

Keith INDUSTRIES
20701 FM 521 ROSHARON, TX 77583
+1-713-991-5670

COATING INSPECTION RECORD

KF-010 FBE 5/27/22 REV2

CUSTOMER: TTB	DATE: 7/7/2026	ITEMS: 6 PCS
CUSTOMER PO: 26H 109149	PAINT SPEC: TPE_26907SP-CONSUMER ENERGY	

SURFACE PREPARATION

ITEM SALT TEST	N/A☐	RESULT: <1	SSPC SP-1 IF NEEDED		DATE	7/7/2026	7/7/2026
ABRASIVE SALT TEST	N/A☐	RESULT: <1	CLEANLINESS REQUIRED	SSPC SP-10	TIME	7:45AM	9:45AM
LABORATORY TESTING	☐YES ☒NO		CLEANLINESS ACHIEVED	☒YES☐NO	AIR TEMP	81°F	90°F
TYPE OF ABRASIVE	STEEL GRIT		PROFILE REQUIRED	2.0-4.5 MILS	RELATIVE HUMIDITY	78%	75%
NOZZLE PSI	☐ N/A 100 PSI ☒		PROFILE ACHIEVED	☒YES☐NO	STEEL	82°F	97°F
BLOTTER TEST	☒PASS		SSPC-VIS 1 USED	☒YES ☐NO	DEW POINT	73°F	81°F
DUST FREE SURFACES	☒YES ☐NO		VISUAL INSPECTION	☒YES ☐NO	STAGE OF OPERATION	PRE-BLAST	PRE-COATING

COATING APPLICATION

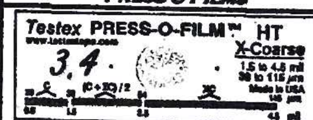
COAT 1ST	PRODUCT NAME: PIPECLAD 2000	BATCH: CH1416SW	MIN MILS: 14	TARGET PRE COAT TEMP: 450 DEG F

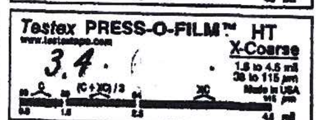
FINAL TESTING

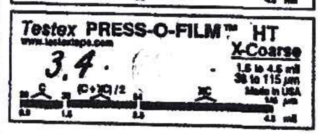
HVST @125V/MIL; 2000V MIN PASS ☒	ADHESION KNIFE TEST: PASS ☒	DFT TEST: PASS ☒	ALL REQUIREMENTS MET: PASS ☒

KI#	DESCRIPTION	MIL AVERAGE
1-1	8" X .322 30DEG BEND	27
1-2	8" X .322 30DEG BEND	26
1-3	8" X .322 30DEG BEND	28
1-4	8" X .322 30DEG BEND	25
2-1	8" X .322 45DEG BEND	27

PRESS O FILMS







KI#	DESCRIPTION	MIL AVERAGE
2-2	8" X .322 45DEG BEND	27

VERIFIED BY KEITH: ____________

Production Bend Hardness Report

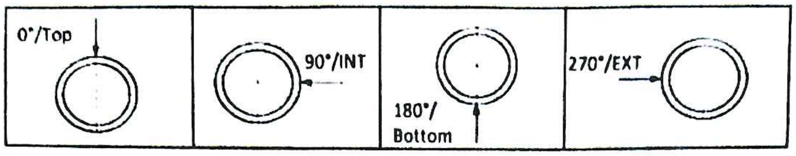
Tulsa Tube Bending

Customer:	DNOW	Report No.:	109149-H-1
Customer PO No.:	662974020	Code of Fabrication:	
TTB Job No.:	26H109149	ASME B16.49 2023	

Mother Pipe Information — **Process Information** — Serial No.: MK8821415112-1754

Pipe Size (in): 8.625 x 0.322 — Grade: API-5L-X52 PSL 2 — Instrument Model No.: UCI-3000H — Probe Serial No.: MK8823417972

Key:



T= Top (Weld if Welded Material)
H= Weld Heat Affected Zone
I= Intrados of Bend
E= Extrados of Bend
TAN= Tangent of Bend
STZ= Start Transition Zone
FTZ= Finish Transition Zone
3/4= 3/4 Point of the Bend Arc

Note 1. Weld & Weld HAZ readings are only required on bends made from welded mother pipe.

Note 2. Readings at 3/4 bend arc locations are only required for production bends at least 30° greater than the QB angle.

Hardness Readings

Min Allowed Values: 137 — Max Allowed Values: 238

Bend ID No.	TAN T 0°	TAN H 0°	TAN 270°	STZ E 270°	STZ I 90°	MID T 0°	MID H 0°	MID E 270°	MID I 90°	FTZ E 270°	FTZ I 90°	3/4 T 0°	3/4 H 0°	3/4 EXT 270°	3/4 INT 90°
109149-1-1	194	189	199	188	182	189	180	181	186	189	185	N/A	N/A	N/A	N/A
109149-1-2	188	179	188	180	190	186	175	182	192	186	189	N/A	N/A	N/A	N/A
109149-1-3	195	188	183	179	180	203	190	182	177	183	178	N/A	N/A	N/A	N/A
109149-1-4	189	181	186	184	181	183	181	190	183	183	184	N/A	N/A	N/A	N/A
109149-2-1	192	187	190	186	184	197	190	183	183	188	187	N/A	N/A	N/A	N/A
109149-2-2	192	187	181	185	183	196	194	185	182	190	186	N/A	N/A	N/A	N/A

READINGS: ☑ MEET ACCEPTANCE CRITERIA — **PERFORMED BY:** JV — **DATE:** 6/12/2026

☐ DO NOT MEET ACCEPTANCE CRITERIA